Introduction to Pathology
&
Cell Injury

LG: 1

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Lecture Outline:

* Introduction to pathology
* Overview of cellular response to injury.
* Causes of Cell Injury

Objectives:

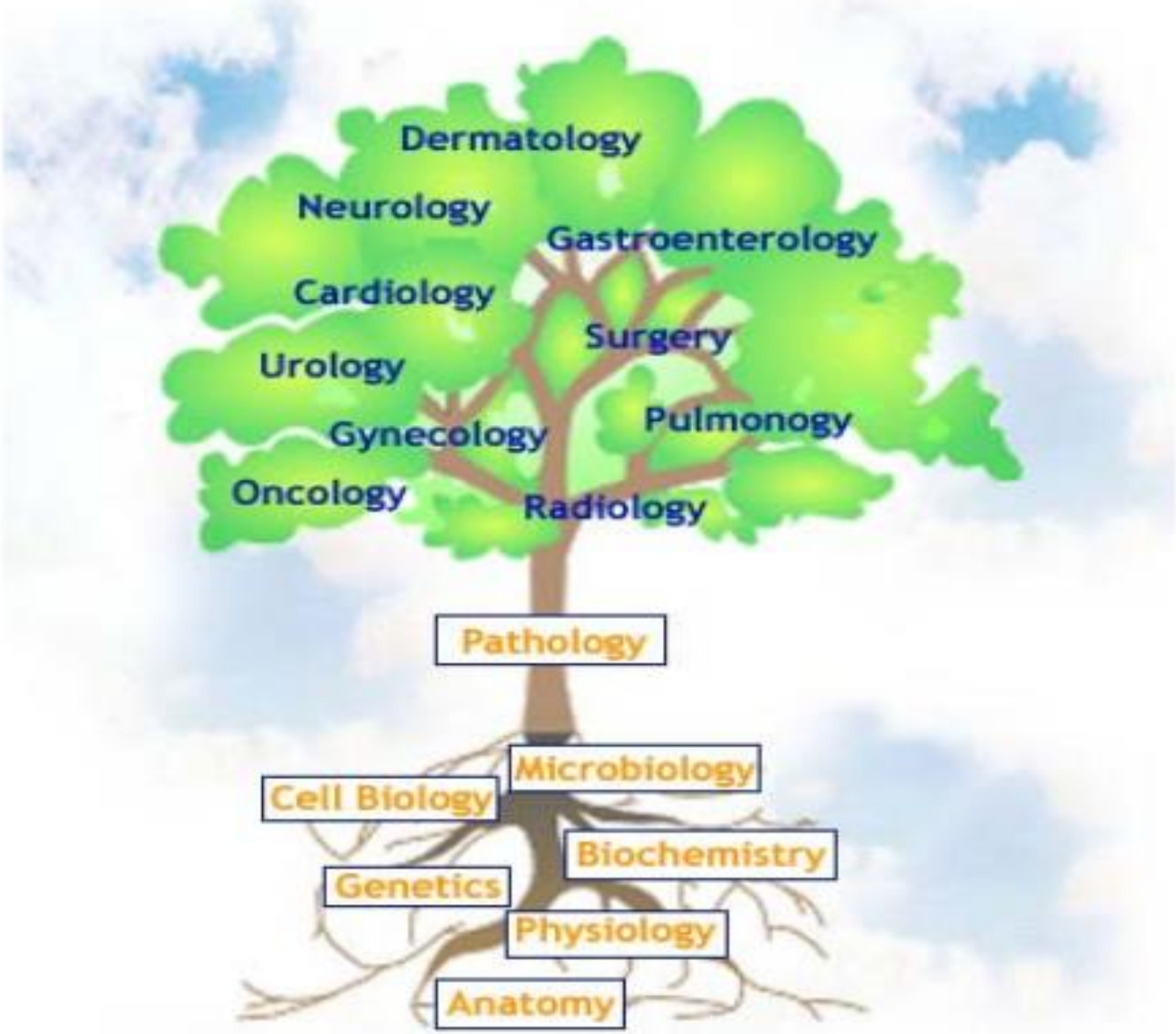
1.To learn the definition of pathology

2. To understand concept of cellular response to stress.

3. To explain causes of cell injury.

- What is pathology?
- Who is a pathologist?
- What is a disease?
- How are diseases diagnosed?

The Tree of Medicine



Pathology (Gr. pathos “disease” + logos “word, reason”)
is the study of the links between diseases and the basic science

What is a Disease?

- **A disease is a physical or functional disorder of normal body systems that places an individual at increased risk of adverse consequences**

- **Diseases are diagnosed by physicians or other health care providers through a combination of tools**

- **When a disease is diagnosed, treatment is given to prevent complications and to improve prognosis**

Diagnosis (Gr. dia “through” + gnosis “knowledge”)

- Diagnoses are made by three general categories of physicians or health care providers:

- **Clinical diagnosticians** identify diseases by examination of patient's history and physical examination

– **Radiologists** identify diseases by imaging the intact body .

– **Pathologists** identify diseases by examining cells and tissues removed from the body

Pathology:

is the study of the structural and functional causes of human disease.

The four aspects of a disease process that form the core of pathology are:

1. The causes of a disease (etiology).

2. The mechanisms of disease development (pathogenesis).

3. The structural alterations induced in cells and tissues by the disease (morphology) including gross and microscopic examination.

4. The functional consequences of the morphologic changes (clinical signs and symptoms).

PATHOLOGY:

- GENERAL
- SYSTEMIC

General Pathology

- It explains the development of basic pathologic mechanisms:
  - **Cell injury.**
  - **Inflammation, repair and healing.**
  - **Hemodynamic (circulatory) disorders.**
  - **Neoplasia.**
  - **Immunopathology**
  - **Genetic disorders**
  - **Infection**

Systemic Pathology

- It explains pathologic mechanisms related to various organ systems:
  - **Cardiovascular System (CVS)**
  - **Respiratory System**
  - **Alimentary System (GIT)**
  - **Hepatobiliary System and Pancreas**
  - **Urinary System**
  - **Male & Female Genital Tract Systems**
  - **Central nervous system (CNS)**
  - **Musculoskeletal system**
  - **Dermatopathology (SKIN)**

Cellular Response To Injury

- **Normal cell function requires a balance between physiologic demands, the structure of cell and metabolic capacity; the result is a steady state, or homeostasis.**

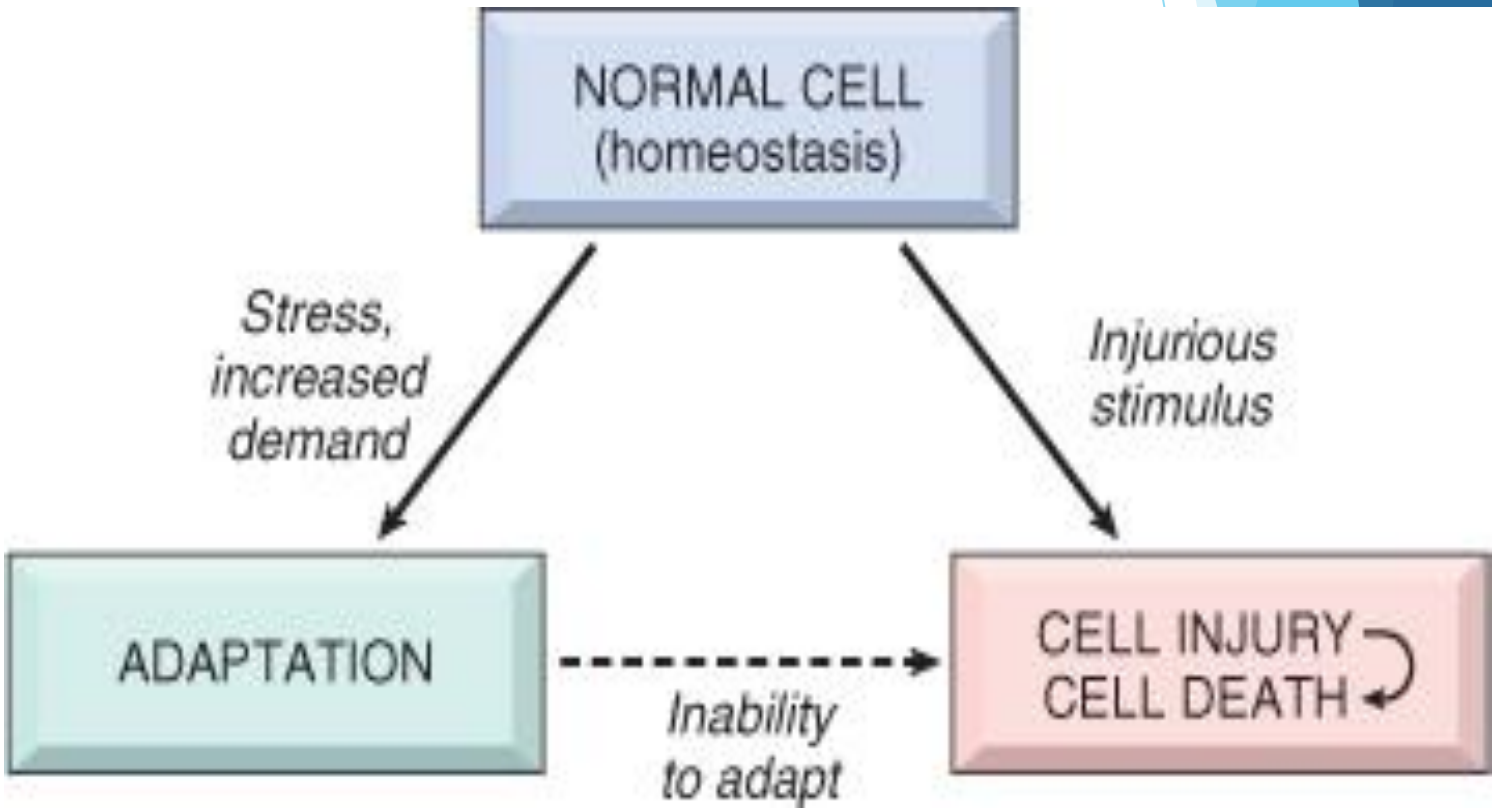
Cells can alter their functional state in response to modest stress to maintain the steady state.

More excessive physiologic stresses or adverse pathologic stimuli (injury), result in:

1. **Adaptation.**

2. **Reversible injury.**

3. **Irreversible injury and cell death.**



Causes of Cell Injury

- Oxygen Deprivation
- Physical Agents
- Chemical Agents and Drugs
- Infectious Agents
- Immunologic Reactions
- Genetic Derangements
- Nutritional Imbalances

Oxygen Deprivation

Which impairs aerobic respiration & the ability to produce ATP.

This is a common cause of cell death.

a. Hypoxia- lack of O2 results in decreased aerobic respiration

b. Ischemia- loss of blood supply

A. Hypoxia

Lack of O2 results in decreased aerobic respiration as in:

- **Reduced blood flow**
- **Inadequate oxygenation of the blood due to cardiorespiratory failure, as in heart failure.**
- **Reduction in the oxygen-carrying capacity of the blood, as in anemia or carbon monoxide (CO) poisoning.**

B. Ischemia:

- Ischemia ("ischemic hypoxia“) Loss of arterial blood flow. ↓ O2 & nutrition to tissues
- Local causes
  - Occlusion of the arteries that bring in fresh blood
  - Occlusion of the veins which allow blood to leave, so that fresh blood can flow in
- Systemic causes
  - Failure of the heart to pump enough blood

- **Ischemia is the commonest cause of hypoxia,**
- **Restoration of blood may lead to recovery OR Ischemia/ Reperfusion injury .**
- **Progressive cell damage**

Ischemia / Reperfusion Injury

- Restoration of blood flow→ influx of high levels of calcium
- Reperfusion increases recruitment of inflammatory cells → **free radical injury**
- Damaged mitochondria induce free radical production & compromise antioxidant defense mechanisms

Reactive Oxygen Species (ROS)

- **Free radicals** are chemical species with a single unpaired electron in an outer orbital.
- Free radicals are extremely **unstable**
- Single free radicals initiate chain reactions which destroy large numbers of organic and inorganic molecules

- **The most important free radicals are probably those derived from oxygen called Reactive oxygen species (ROS)**
- **Ex: superoxide (O.-2);**

  hydroxyl radical (OH.);

  hydrogen peroxide ((H2O2) and

  Nitric oxide (NO)

- **The different types of ROS are produced by two major pathways:**

- ✓ **ROS are produced normally during the reduction-oxidation (redox) reactions**
- ✓ **ROS are produced in phagocytic leukocytes, mainly neutrophils and macrophages**

- **Free radicals may be generated in the following ways:**

1.The absorption of radiant energy (e.g., ultraviolet light, x-rays). Ionizing radiation can hydrolyze water into hydroxyl (• OH) and hydrogen (H •) free radicals.

2.The enzymatic metabolism of exogenous chemicals (e.g., carbon tetrachloride)

3. Inflammation, in which free radicals are produced by leukocytes

Free Radical Defense

- 1- Spontaneous decay.
- 2- Antioxidants
- 3- Free radical scavenging systems.

Free-radical effects:

1.Oxidation of fatty acids in membranes ("lipid peroxidation").

2.Cross-linking of proteins.(protein denaturation)

3.Genetic mutations (DNA depolymerization)

Physical Agents

- Mechanical trauma
- Extremes of temperature (burns, deep cold)
- Radiation
- Electric shock

Chemical Agents and Drugs

- Drugs & hypertonic concentration of salt **(deranging electrolyte homeostasis)**
- Poisons **(arsenic, cyanide, or mercuric salts)**
- Insecticides and Herbicides
- Air pollutant **(carbon monoxide)**
- Occupational hazard **(asbestos)**
- Alcohol and Narcotic drugs

Infectious Agents

- Parasites
- Fungi
- Bacteria
- Rickettsiae
- Viruses

Immunologic Reactions

- Hypersensetivity (Anaphylactic) reaction to foreign protein or drug
- Reactions to endogenous self-antigens –autoimmune diseases

Nutritional Imbalances

- Protein-calorie deficiencies
- Vitamin deficiencies
- Anorexia nervosa
- Excesses of lipids (Obesity, Atherosclerosis)
- Metabolic diseases (Diabetes)

Genetics Derangements

- Congenital malformation (Down's syndrome)
- Decreased life of red blood cell (Thalassemia, Sickle cell anaemia))
- Inborn errors of metabolism

References:

1. Robbins and Cotran

PATHOLOGIC BASIS OF DISEASES

By : KUMAR – ABBAS – ASTER

2. Robbins

BASIC PATHOLOGY

By: KUMAR- ABBAS – ASTER

Thank You